

Dhanvardini Rajendran

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Summary

Data and application engineer with 2+ years building cloud-native data pipelines, **KPI dashboards**, and **process improvement** automation on AWS and Azure. Eager to bring that cloud engineering foundation to GlobalFoundries and help modernize tapeout workflows through scalable tooling and data-driven optimization.

Education

Northeastern University, Boston MS in Information Systems GPA: 3.7/4.0	Aug 2024 – May 2026
Courses: Generative AI w/ LLMs in Data Engineering, Advanced Data Architecture & Business Intelligence	
SASTRA University B.Tech in Computer Science	June 2018 – June 2022

Professional Experience

Optum, UnitedHealth Group Software Engineer	Feb 2023 – Aug 2024
- Designed and delivered 10+ ETL pipelines on Azure Databricks, processing 4B+ records across multi-frequency schedules and publishing enriched data products consumed by 5+ downstream teams. - Reduced pipeline failure rate from 40% to 10% through root cause analysis, SQL query optimization, and embedded data quality gates including reconciliation, deduplication, and null validation. - Built and distributed recurring KPI dashboards in Power BI and Tableau, surfacing performance trends and anomalies for leadership on weekly and monthly cadences. - Productionized an enterprise analytics platform, engineering 5+ production-grade PySpark pipelines with window functions, deduplication, and multi-table joins from a POC handoff.	
Optum, UnitedHealth Group Associate Software Engineer	July 2022 – Feb 2023
- Enhanced internal execution framework with step sequencing, reconciliation, and automated recovery, enabling deterministic pipeline restarts and reducing manual intervention. - Deployed data assets across multiple Azure Databricks workspaces using Unity Catalog, implementing fine-grained access control and three-tier namespace governance. - Implemented SQL performance optimizations including predicate pushdown and selective aggregation on billion-row datasets stored in Delta Lake on ADLS.	
Optum, UnitedHealth Group Software Engineer Intern	Jan 2022 – May 2022
- Built Gold-layer data products on Azure Databricks using PySpark and SQL, publishing production Delta Tables for consumption by analytics and reporting teams. - Engineered Azure Data Factory orchestration pipelines with dependency-aware scheduling across multiple stages, managing end-to-end data flow through ADLS.	

Technical Projects

IMDb Analytics: AWS Cloud-Native ETL & Redshift Warehouse Pipeline	Dec 2025
Designed a Redshift Serverless star schema with 5 conformed dimensions, defined distribution and sort keys for query optimization, profiled 7 datasets using Alteryx, and built Amazon QuickSight dashboards visualizing performance metrics.	
PodcastIQ: AI-Powered Multi-Agent Semantic Search Platform	March 2026
Architected a 4-layer Snowflake warehouse with dbt Cloud, built modular models with reusable macros, enforced data quality via schema tests in GitHub Actions CI/CD, and orchestrated a 9-agent LangGraph system achieving 95.8% accuracy.	
META TradePulse: Real-Time Quantitative Trading Signal Platform	Nov 2025
Engineered a real-time pipeline with Dockerized Kafka producer, PySpark Structured Streaming computing technical indicators, Snowflake persistence via Streams and Tasks, and interactive alerts via Streamlit dashboard.	

Awards & Certifications

Certifications: AWS Certified Data Engineer, AWS Certified Solutions Architect, Azure Certified Data Fundamentals
Laurel & Scroll 100: Northeastern University's top 100 graduating students for academic excellence and leadership
COE Award: College of Engineering Outstanding Graduate Student Award, Northeastern University
AWS All Builders Welcome Grant: Selected as 1 of 100 globally for technical innovation at AWS re:Invent

Technical Skills

Languages & Frameworks: Python, SQL, Angular, PySpark, Streamlit, LangGraph, LangChain
Cloud & Data Platforms: AWS (S3, Glue, Redshift, **QuickSight**), Azure (Data Factory, Databricks, ADLS Gen2), Snowflake
Databases & Warehousing: MSSQL, SQL Server, Delta Lake, Redshift, Snowflake, PostgreSQL
Visualization & BI: **QuickSight**, Power BI, Tableau, Streamlit, Dashboard Development
DevOps & Tools: Git, Jenkins, Azure DevOps, GitHub Actions, Docker, **JIRA**, Confluence